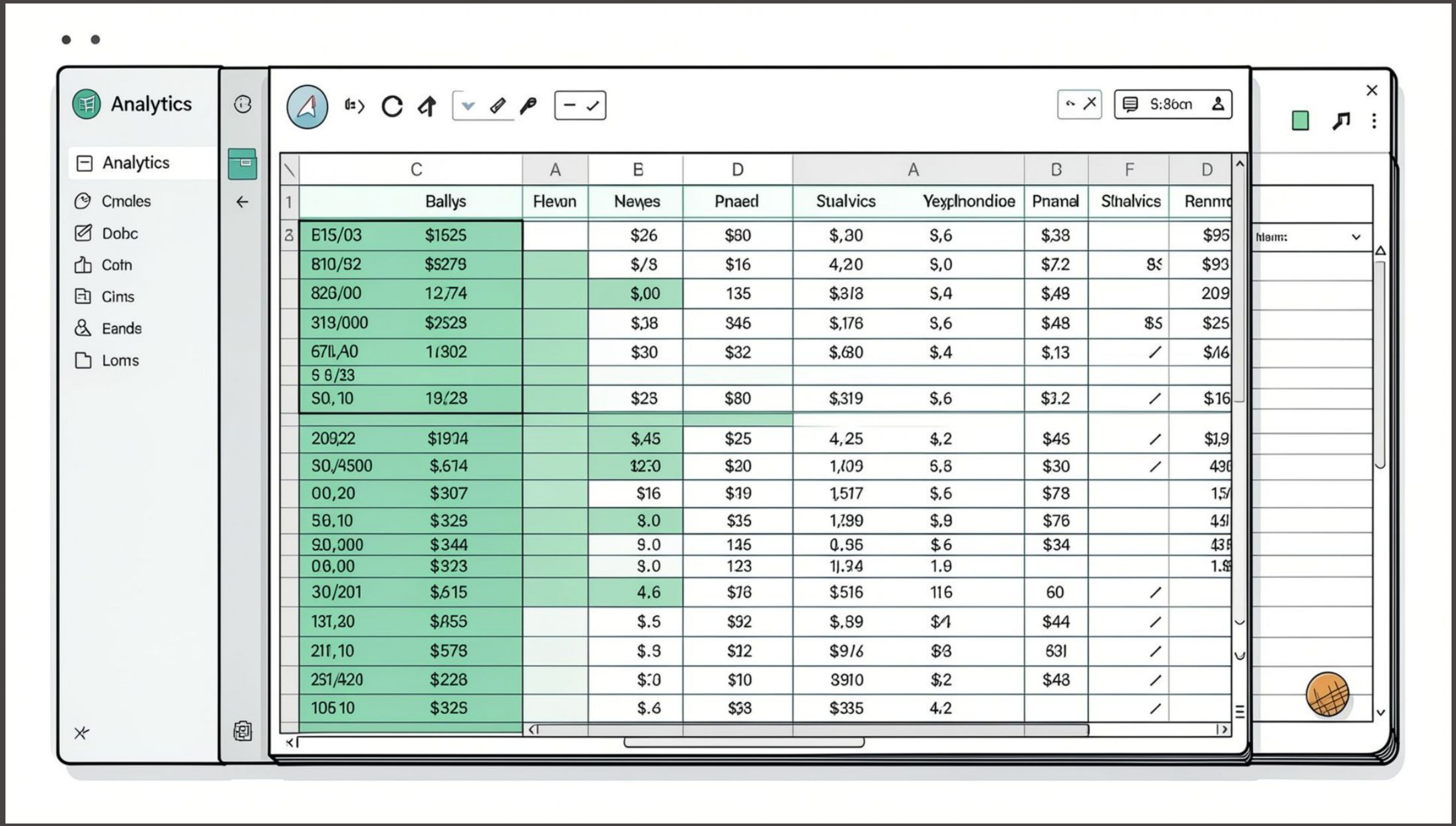
Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — spending patterns, customer segments, and subscription behavior to guide strategic business decisions.

Dataset at a Glance



Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount, Frequency, Review Rating, Shipping Type).

3,900 Rows

Individual purchase transactions

18 Columns

Rich feature set

37 Missing

Values in Review Rating

Data Preparation in Python

Load & Explore

Imported via pandas; used `df.info()` and `.describe()` for initial structure and statistics.

Clean & Standardize

Imputed missing Review Ratings using median per category; renamed columns to snake_case.

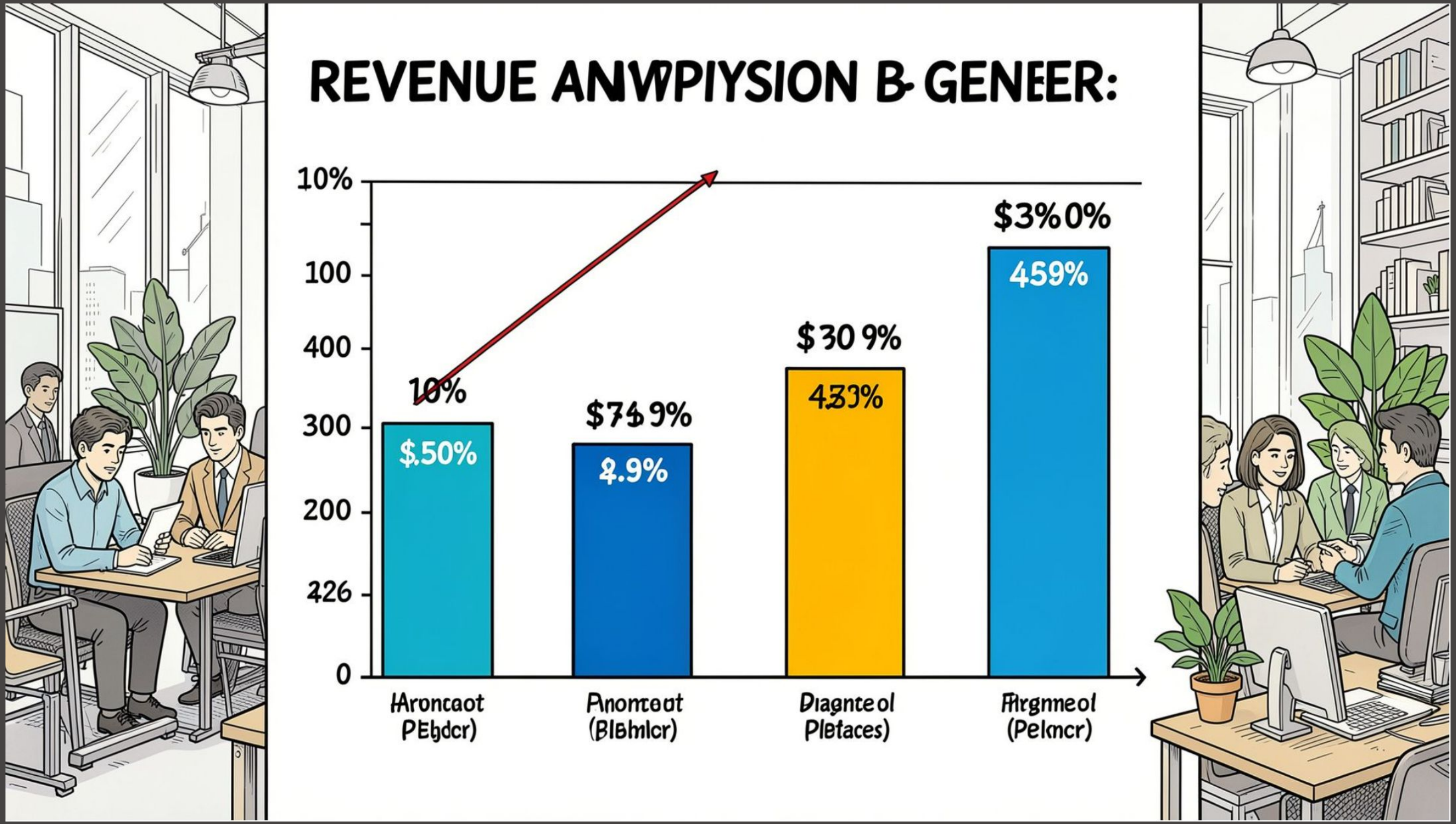
Feature Engineering

Created `age_group` by binning ages and `purchase_frequency_days` from purchase data.

Database Integration

Connected to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.

Revenue & Spending Insights



Revenue by Gender

Compared total revenue generated by male vs. female customers.

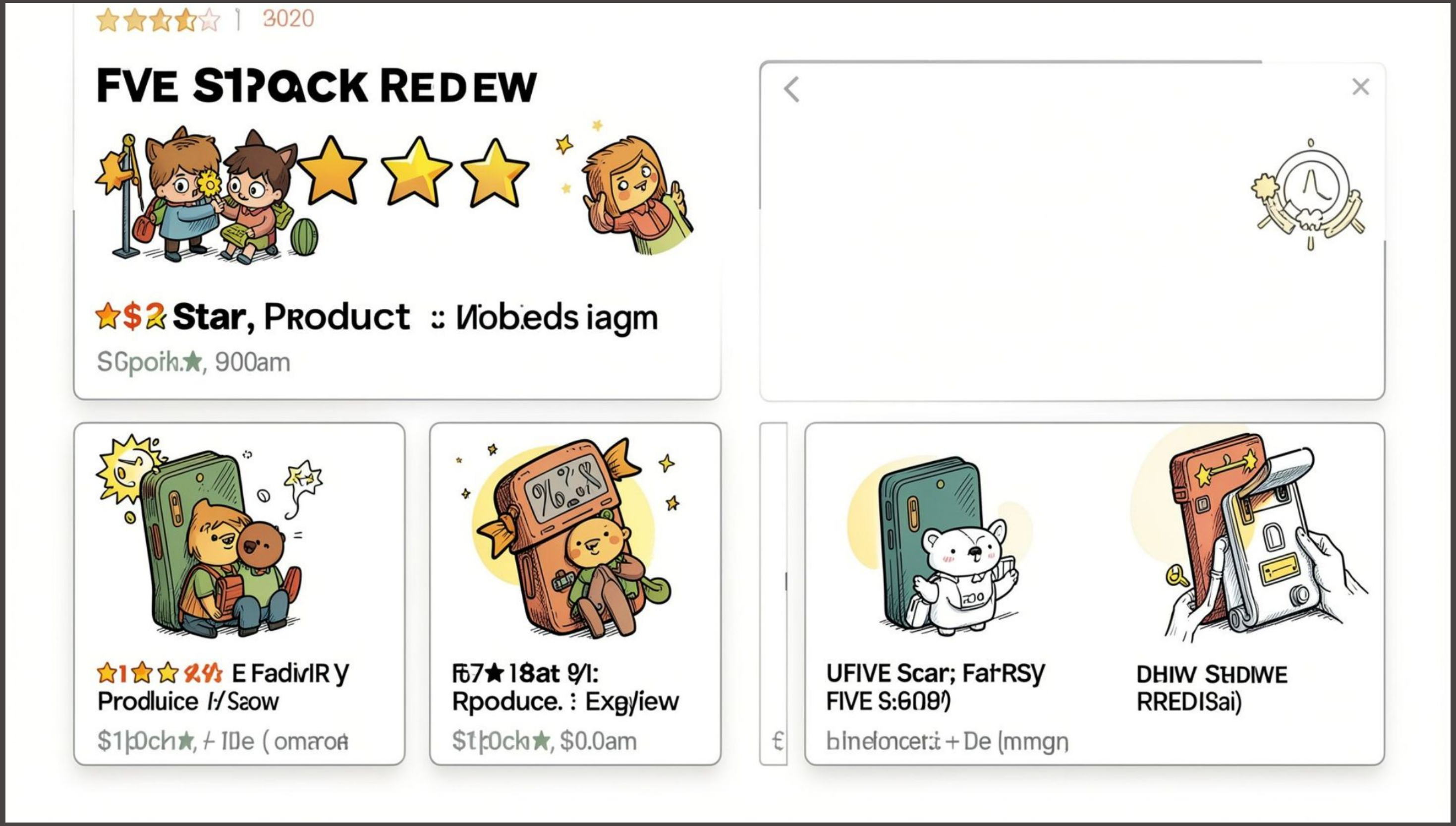
High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

Revenue by Age Group

Calculated total revenue contribution of each age group.

Products & Ratings



Top 5 Products by Rating

Found products with the highest average review ratings.

Top 3 Products per Category

Listed the most purchased products within each category.

Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

Products Purchased by 100+ Customers

Identified which products have been purchased by more than 100 customers.

Shipping, Payments & Categories

Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

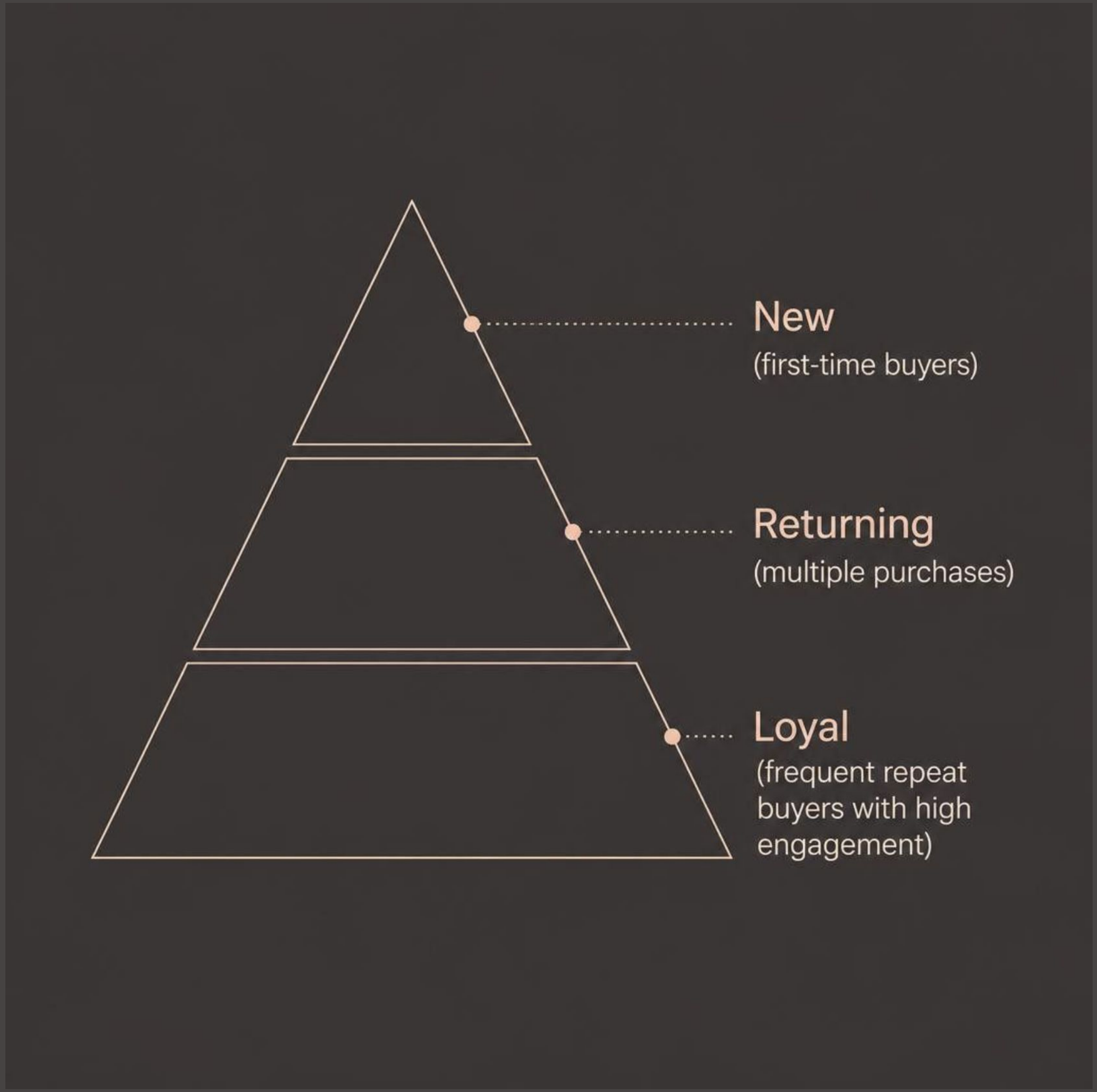
Payment Method Analysis

Which payment method generates the highest average purchase amount?

Category Purchase Analysis

Average purchase amount per product category — which categories exceed $50?

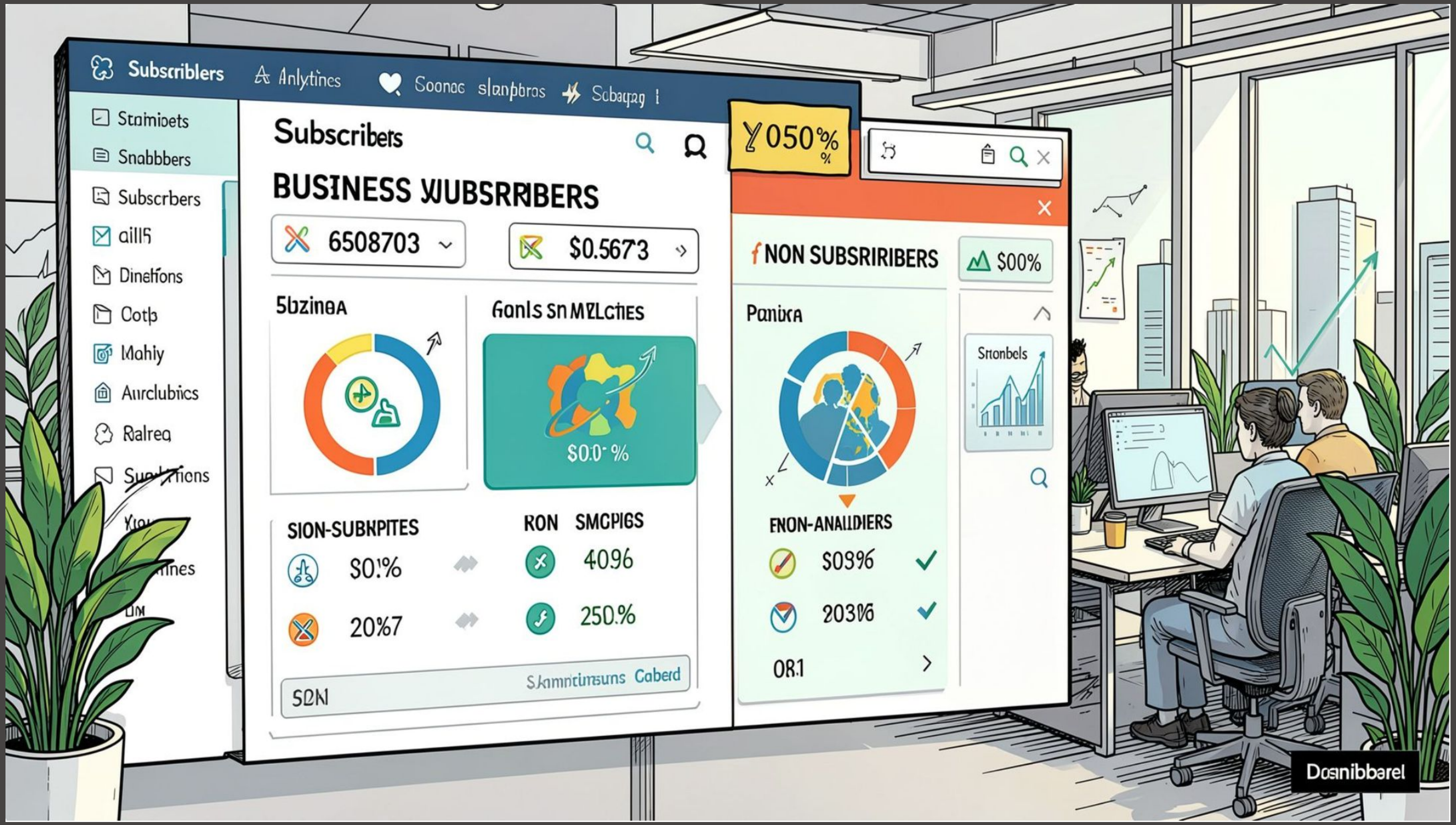
Customer Segmentation



Customers were classified into **New**, **Returning**, and **Loyal** segments based on purchase history. Repeat buyers with more than 5 purchases were analyzed to check whether they are more likely to subscribe.

Repeat buyers with **>5 purchases** were checked for subscription likelihood — a key lever for loyalty programs.

Subscriber & Location Analysis



Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

Location Analysis

Which locations have the highest number of customers and what is their total revenue?

Power BI Dashboard

An **interactive Power BI dashboard** was built to present all insights visually — enabling stakeholders to explore revenue, segments, product performance, and subscription trends at a glance.

Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers.

Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control.

Product Positioning

Highlight top-rated and best-selling products in campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.